Multi-Rail Isolated Power Supply Monitor
System Test Report

Overall system status	PASS	**Checklist file**	capstone_system_test_checklist.csv
Rail measurement status	PASS	**Raw measurement data**	capstone_rail_test_raw.csv
DUT	1	**Operator(s)**	MC
Latest-attempt date range	2026-10-18T18:34:42 to 2026-10-18T18:48:59	**IOC prefix**	TEST:GIGA
Electronic load	RIGOL TECHNOLOGIES, DL3021, DL3A284M00251	**Report generated**	2026-10-18T18:53:22

Acceptance criteria used

- **Voltage accuracy:** IOC voltage must agree with the external reference within ±2% across the tested operating range.
- **Current accuracy:** IOC current must agree with the external reference within ±5%. At low current, the larger of the percentage tolerance or an absolute floor is used: 25 mA for 39 mOhm channels and 20 mA for 50 mOhm channels.
- **Polarity:** Positive and negative rail voltages must have the correct sign. Current follows the firmware convention for all rails.
- **Display / EPICS agreement:** Local-display and EPICS values must represent the same underlying measurement within displayed resolution or rounding.
- **Limits and persistence:** Programmed limits must assert and clear correctly, and persistent settings must survive a power cycle.
- **Network recovery:** EPICS communication must resume normally after Ethernet communication is interrupted and restored.
- **Automated verification:** Rail accuracy, Channel Access connectivity, PV updates, and other electronically observable checks are derived automatically from the latest rail-test results.
- **Operator verification:** During each rail test, the operator is prompted only for local-display observations that cannot be measured automatically. These observations are reused for the applicable system-test checklist items.

System test matrix — latest result

ID	Test	Result	Attempt	Source / notes
ST-01	Power On and Initialization	PASS	1	4 auto + 6 checklist: All checklist steps are satisfied by automatic evidence and/or operator confirmation.
ST-02	Rail Identification	PASS	—	5 auto: All checklist steps are satisfied by automatic evidence and/or operator confirmation.
ST-03	Voltage Accuracy	PASS	—	2 auto: All checklist steps are satisfied by automatic evidence and/or operator confirmation.
ST-04	Current Accuracy	PASS	—	2 auto: All checklist steps are satisfied by automatic evidence and/or operator confirmation.
ST-05	Dynamic Load Response	PASS	—	5 auto: All checklist steps are satisfied by automatic evidence and/or operator confirmation.
ST-06	Local Display	PASS	—	5 auto: All checklist steps are satisfied by automatic evidence and/or operator confirmation.
ST-07	EPICS Communications	PASS	—	8 auto: All checklist steps are satisfied by automatic evidence and/or operator confirmation.
ST-08	Limit / Alarm Operation	PASS	1	10 checklist: All checklist steps are satisfied by automatic evidence and/or operator confirmation.
ST-09	Persistent Settings	PASS	1	8 checklist: All checklist steps are satisfied by automatic evidence and/or operator confirmation.
ST-10	Communication Recovery	PASS	1	2 auto + 6 checklist: All checklist steps are satisfied by automatic evidence and/or operator confirmation.
ST-11	Repeatability	PASS	1	3 auto + 2 checklist: All checklist steps are satisfied by automatic evidence and/or operator confirmation.
ST-12	End-To-End Demonstration	PASS	1	5 auto + 2 checklist: All checklist steps are satisfied by automatic evidence and/or operator confirmation.

Disposition by rail (latest attempt only)

Rail	Attempt	IOC channel	Status	Start	End
+5VA	1	6VA	PASS	2026-10-18 18:34:42	2026-10-18 18:35:10
+5VB	2	6VB	PASS	2026-10-18 18:37:29	2026-10-18 18:37:57
+5V	1	P5V	PASS	2026-10-18 18:39:27	2026-10-18 18:39:55
-5V	1	M5V	PASS	2026-10-18 18:40:59	2026-10-18 18:41:28
+15V	1	P15V	PASS	2026-10-18 18:43:04	2026-10-18 18:43:32
-15V	3	M15V	PASS	2026-10-18 18:46:38	2026-10-18 18:47:06

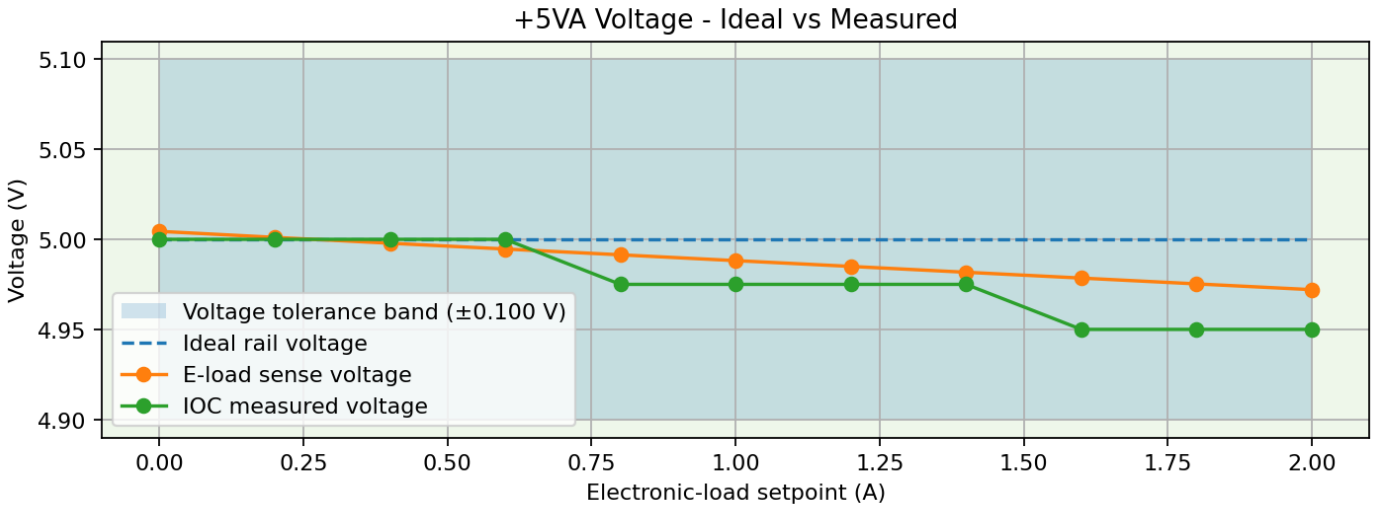
Local-display observations captured during rail testing

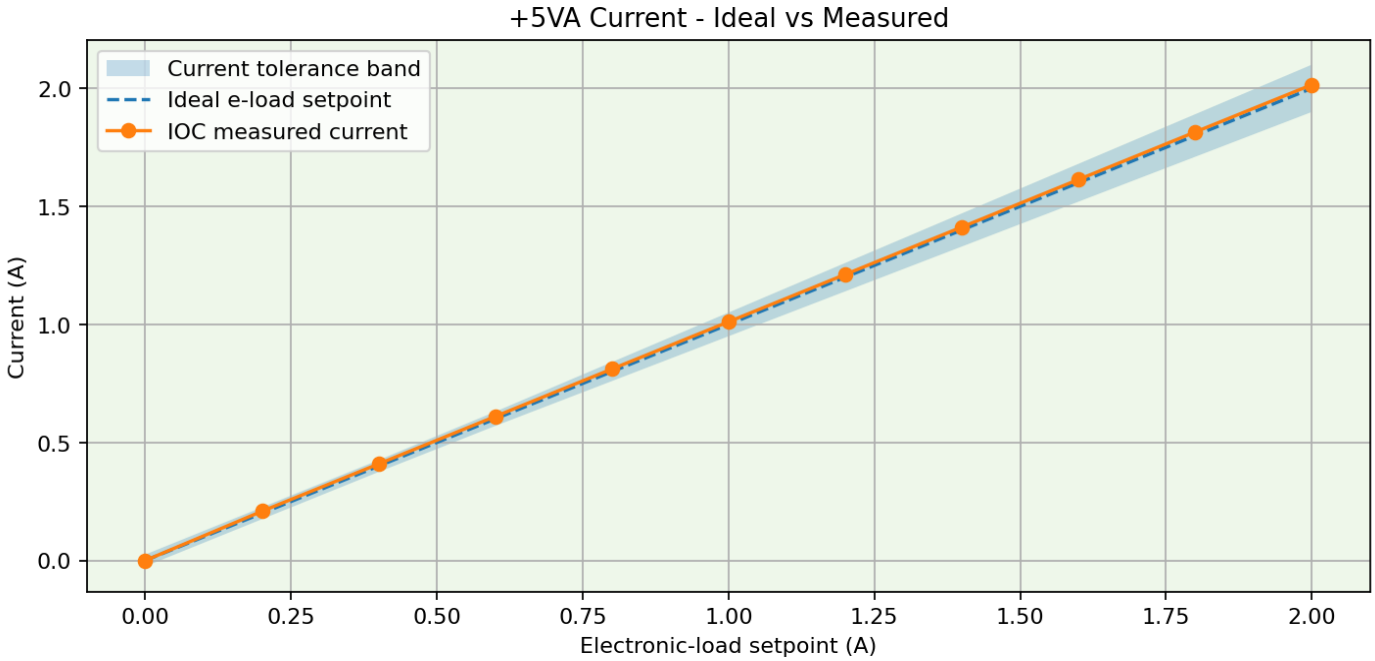
Rail	Identity / polarity	Updates with load	Agrees with EPICS	Limits / status	Network state
+5VA	PASS	PASS	PASS	PASS	PASS
+5VB	PASS	PASS	PASS	PASS	PASS
+5V	PASS	PASS	PASS	PASS	PASS
-5V	PASS	PASS	PASS	PASS	PASS
+15V	PASS	PASS	PASS	PASS	PASS
-15V	PASS	PASS	PASS	PASS	PASS

+5VA detailed results - latest attempt

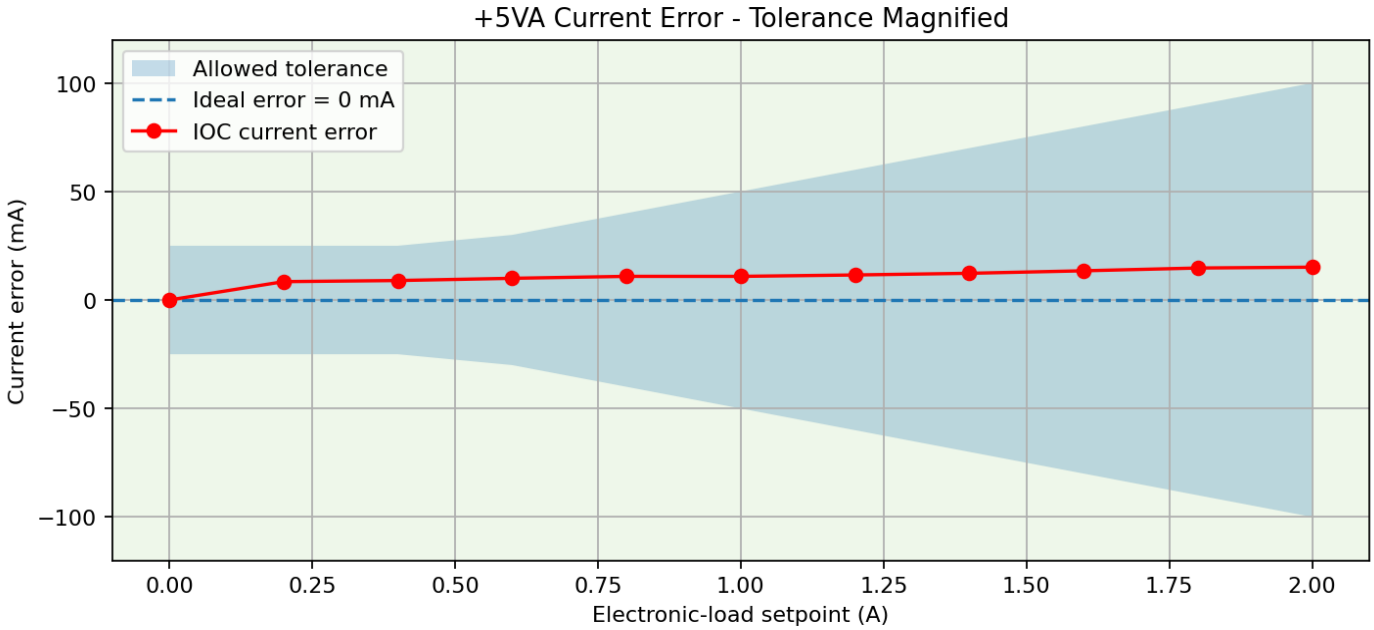
Attempt 1 — PASS IOC channel: 6VA Start: 2026-10-18T18:34:42 End: 2026-10-18T18:35:10

Ideal versus measured response





Current error detail

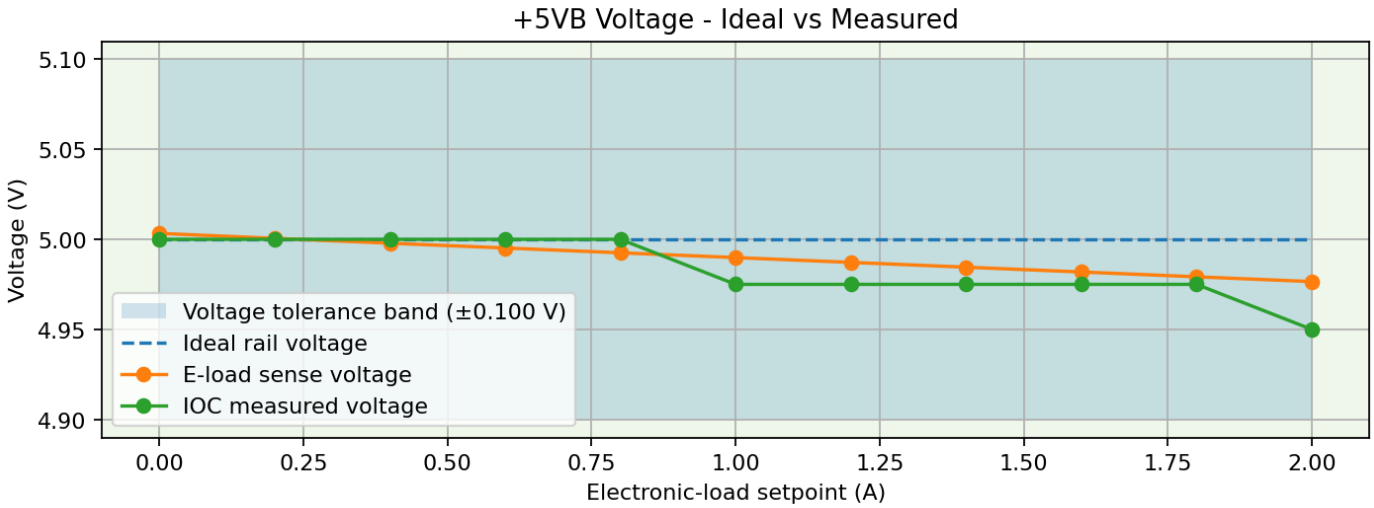


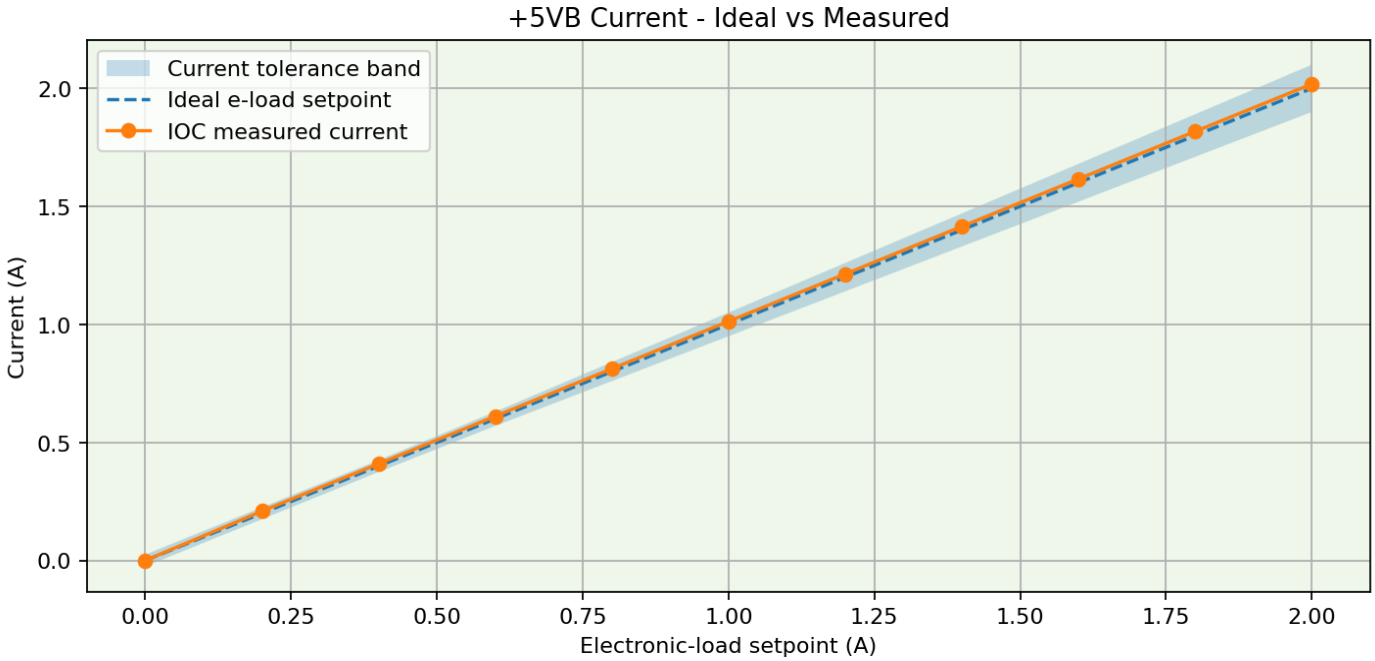
Set I (A)	Rigol V (V)	V ref (V)	IOC V (V)	V err (V)	V tol (V)	Rigol I (A)	IOC I (A)	I err (A)	I tol (A)	Result
0.0000	5.0044	+5.0044	+5.0000	-0.0044	0.1001	0.0000	0.0000	+0.0000	0.0250	PASS
0.2000	5.0011	+5.0011	+5.0000	-0.0011	0.1000	0.1995	0.2085	+0.0090	0.0250	PASS
0.4000	4.9978	+4.9978	+5.0000	+0.0022	0.1000	0.3991	0.4090	+0.0099	0.0250	PASS
0.6000	4.9946	+4.9946	+5.0000	+0.0054	0.1000	0.5999	0.6100	+0.0101	0.0300	PASS
0.8000	4.9914	+4.9914	+4.9750	-0.0164	0.1000	0.8003	0.8109	+0.0106	0.0400	PASS
1.0000	4.9882	+4.9882	+4.9750	-0.0132	0.1000	0.9997	1.0109	+0.0112	0.0500	PASS
1.2000	4.9850	+4.9850	+4.9750	-0.0100	0.1000	1.1993	1.2115	+0.0123	0.0600	PASS
1.4000	4.9817	+4.9817	+4.9750	-0.0067	0.1000	1.3995	1.4123	+0.0128	0.0700	PASS
1.6000	4.9785	+4.9785	+4.9500	-0.0285	0.1000	1.5999	1.6135	+0.0135	0.0800	PASS
1.8000	4.9753	+4.9753	+4.9500	-0.0253	0.1000	1.8002	1.8147	+0.0145	0.0900	PASS
2.0000	4.9721	+4.9721	+4.9500	-0.0221	0.1000	1.9997	2.0151	+0.0154	0.1000	PASS

+5VB detailed results - latest attempt

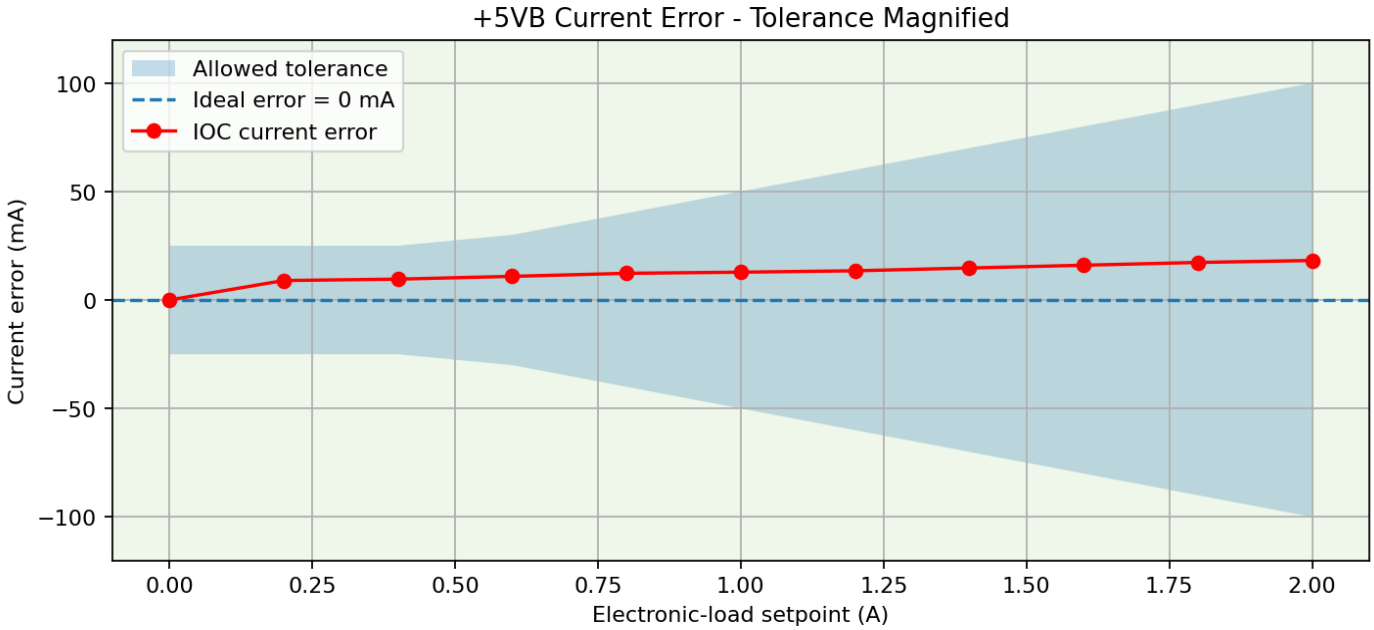
Attempt 2 — PASS IOC channel: 6VB Start: 2026-10-18T18:37:29 End: 2026-10-18T18:37:57

Ideal versus measured response





Current error detail

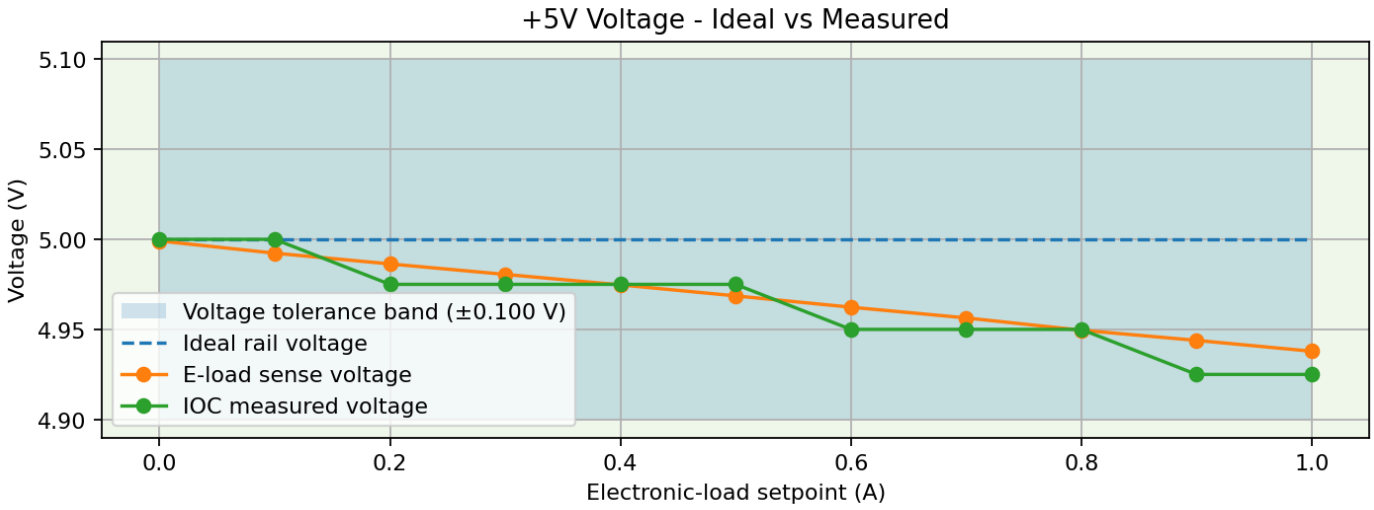


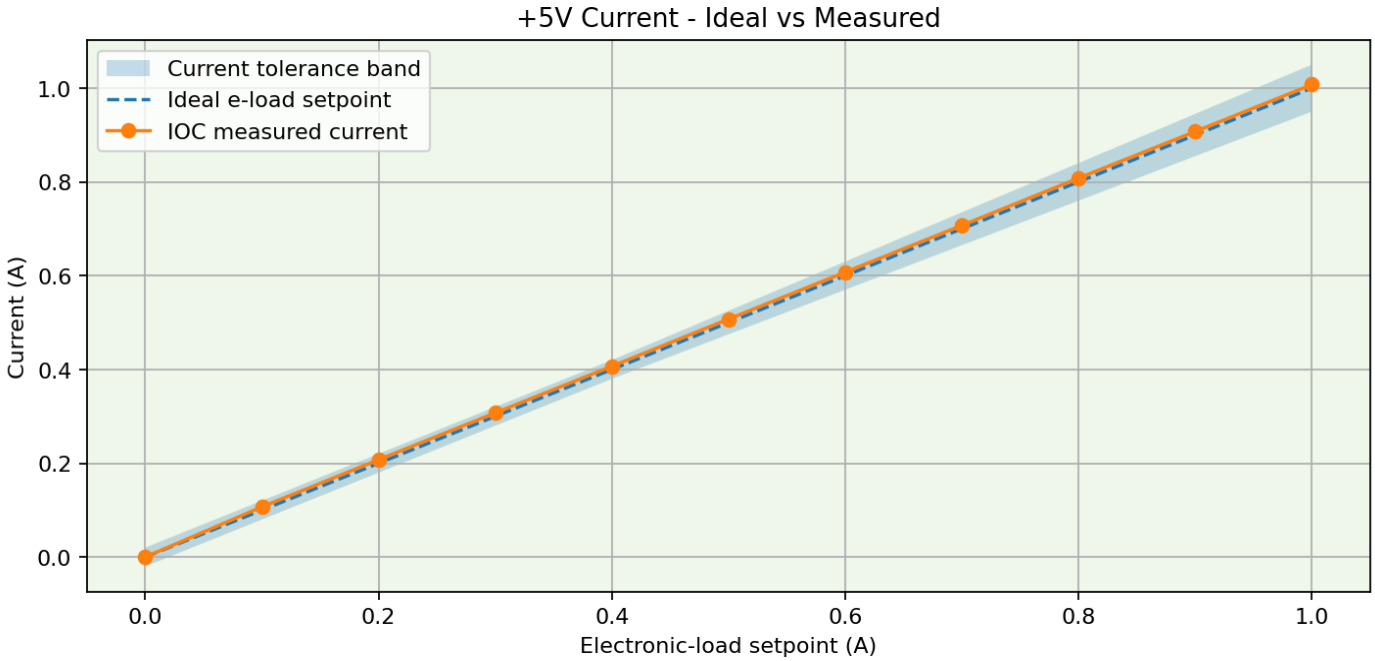
Set I (A)	Rigol V (V)	V ref (V)	IOC V (V)	V err (V)	V tol (V)	Rigol I (A)	IOC I (A)	I err (A)	I tol (A)	Result
0.0000	5.0033	+5.0033	+5.0000	-0.0033	0.1001	0.0000	0.0000	+0.0000	0.0250	PASS
0.2000	5.0005	+5.0005	+5.0000	-0.0005	0.1000	0.1993	0.2090	+0.0096	0.0250	PASS
0.4000	4.9979	+4.9979	+5.0000	+0.0021	0.1000	0.3990	0.4096	+0.0106	0.0250	PASS
0.6000	4.9952	+4.9952	+5.0000	+0.0048	0.1000	0.5997	0.6109	+0.0112	0.0300	PASS
0.8000	4.9925	+4.9925	+5.0000	+0.0075	0.1000	0.8002	0.8123	+0.0121	0.0400	PASS
1.0000	4.9899	+4.9899	+4.9750	-0.0149	0.1000	0.9995	1.0128	+0.0133	0.0500	PASS
1.2000	4.9872	+4.9872	+4.9750	-0.0122	0.1000	1.1991	1.2135	+0.0143	0.0600	PASS
1.4000	4.9845	+4.9845	+4.9750	-0.0095	0.1000	1.3994	1.4147	+0.0153	0.0700	PASS
1.6000	4.9819	+4.9819	+4.9750	-0.0069	0.1000	1.5998	1.6160	+0.0162	0.0800	PASS
1.8000	4.9792	+4.9792	+4.9750	-0.0042	0.1000	1.8002	1.8173	+0.0171	0.0900	PASS
2.0000	4.9765	+4.9765	+4.9500	-0.0265	0.1000	1.9997	2.0182	+0.0185	0.1000	PASS

+5V detailed results - latest attempt

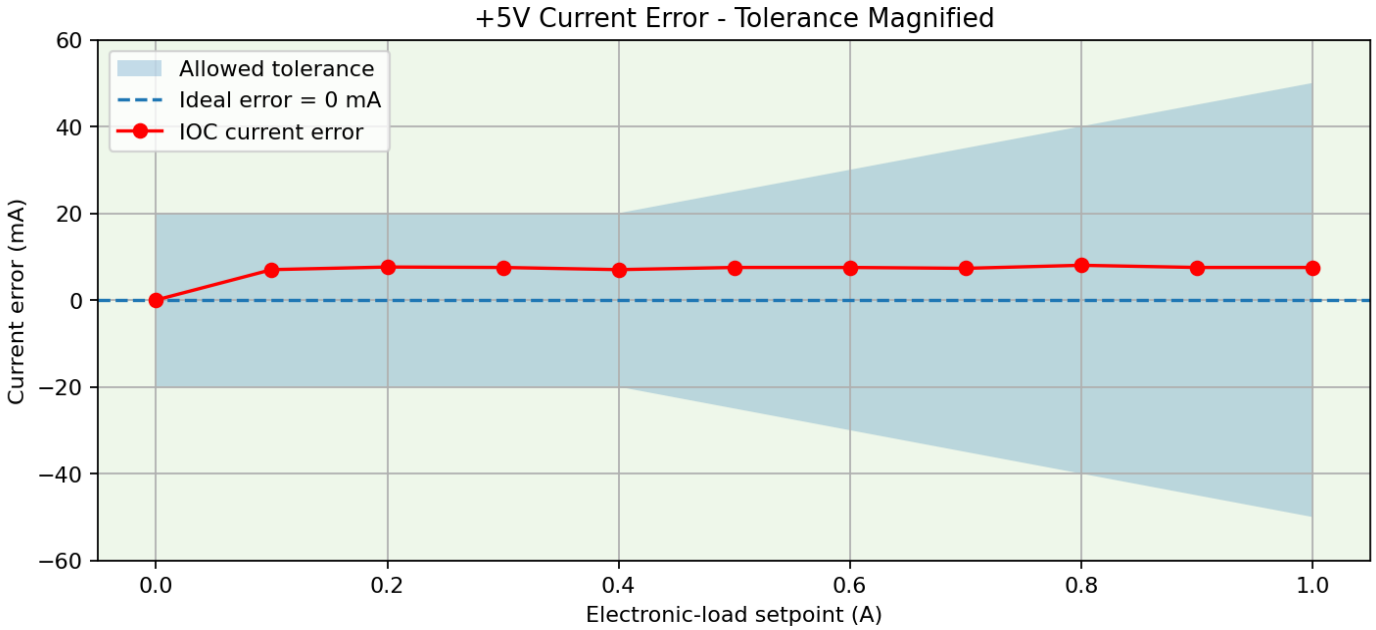
Attempt 1 — PASS IOC channel: P5V Start: 2026-10-18T18:39:27 End: 2026-10-18T18:39:55

Ideal versus measured response





Current error detail

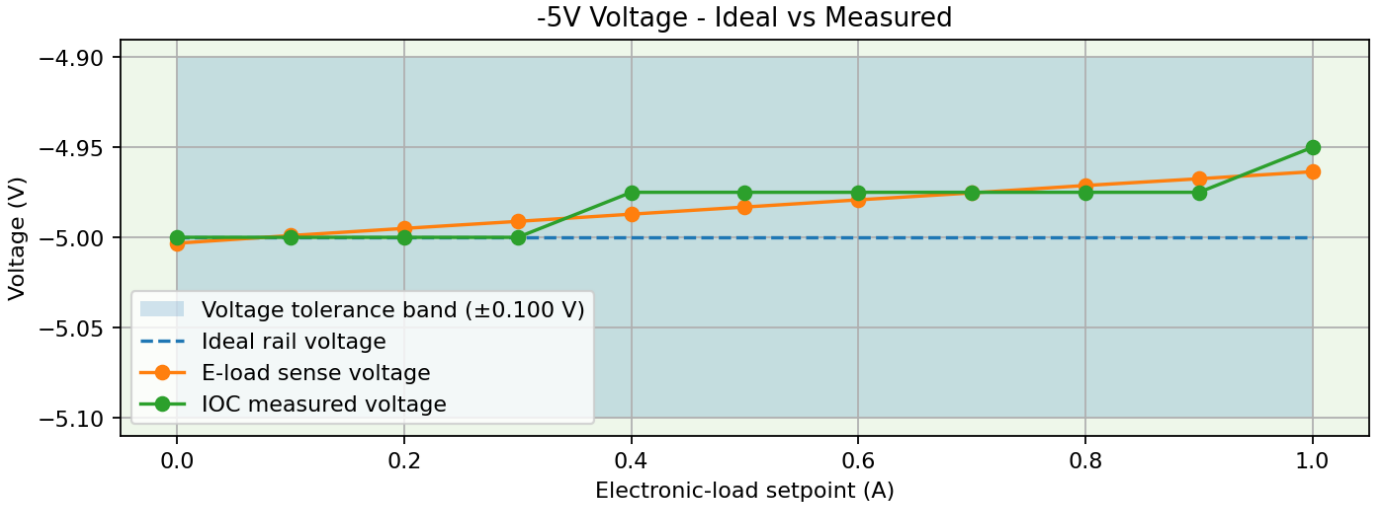


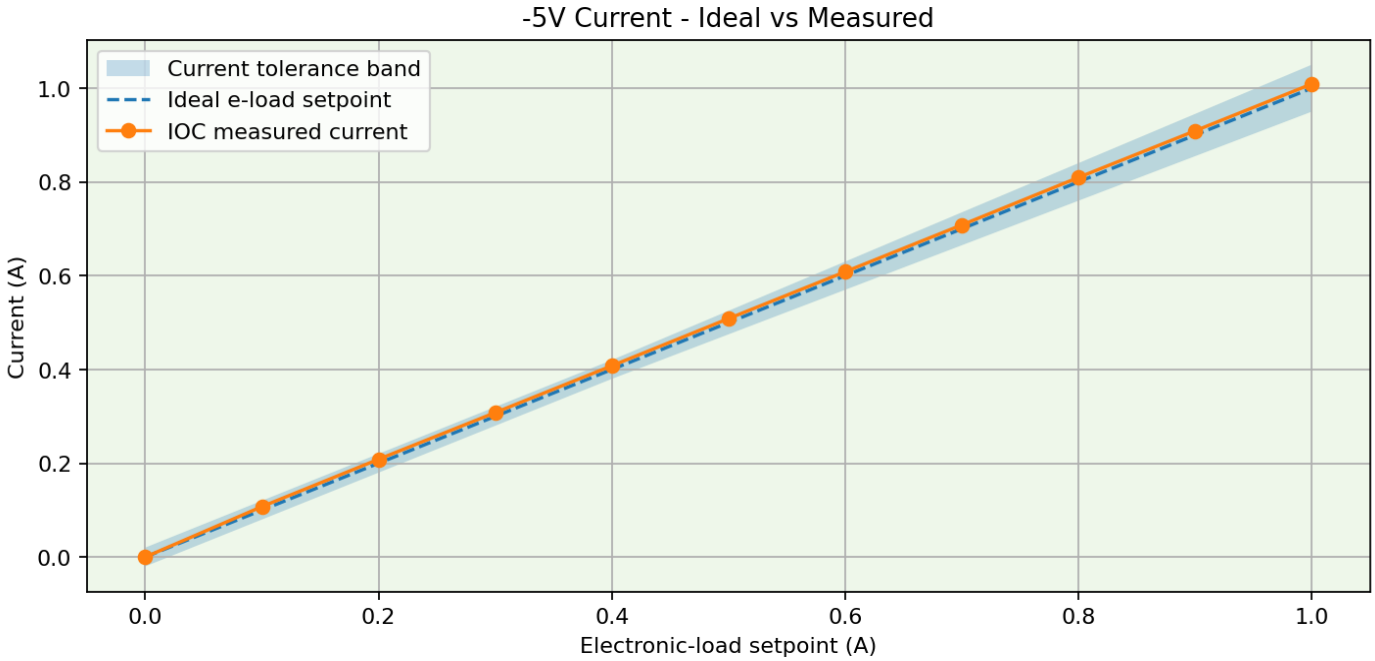
Set I (A)	Rigol V (V)	V ref (V)	IOC V (V)	V err (V)	V tol (V)	Rigol I (A)	IOC I (A)	I err (A)	I tol (A)	Result
0.0000	4.9991	+4.9991	+5.0000	+0.0009	0.1000	0.0000	0.0000	+0.0000	0.0200	PASS
0.1000	4.9923	+4.9923	+5.0000	+0.0077	0.1000	0.0986	0.1070	+0.0084	0.0200	PASS
0.2000	4.9863	+4.9863	+4.9750	-0.0113	0.1000	0.1995	0.2076	+0.0081	0.0200	PASS
0.3000	4.9805	+4.9805	+4.9750	-0.0055	0.1000	0.2992	0.3075	+0.0083	0.0200	PASS
0.4000	4.9747	+4.9747	+4.9750	+0.0003	0.1000	0.3991	0.4070	+0.0079	0.0200	PASS
0.5000	4.9686	+4.9686	+4.9750	+0.0064	0.1000	0.4996	0.5075	+0.0079	0.0250	PASS
0.6000	4.9623	+4.9623	+4.9500	-0.0123	0.1000	0.5999	0.6075	+0.0076	0.0300	PASS
0.7000	4.9564	+4.9564	+4.9500	-0.0064	0.1000	0.6996	0.7073	+0.0077	0.0350	PASS
0.8000	4.9496	+4.9496	+4.9500	+0.0004	0.1000	0.8003	0.8080	+0.0077	0.0400	PASS
0.9000	4.9439	+4.9439	+4.9250	-0.0189	0.1000	0.8998	0.9075	+0.0077	0.0450	PASS
1.0000	4.9379	+4.9379	+4.9250	-0.0129	0.1000	0.9996	1.0075	+0.0079	0.0500	PASS

-5V detailed results - latest attempt

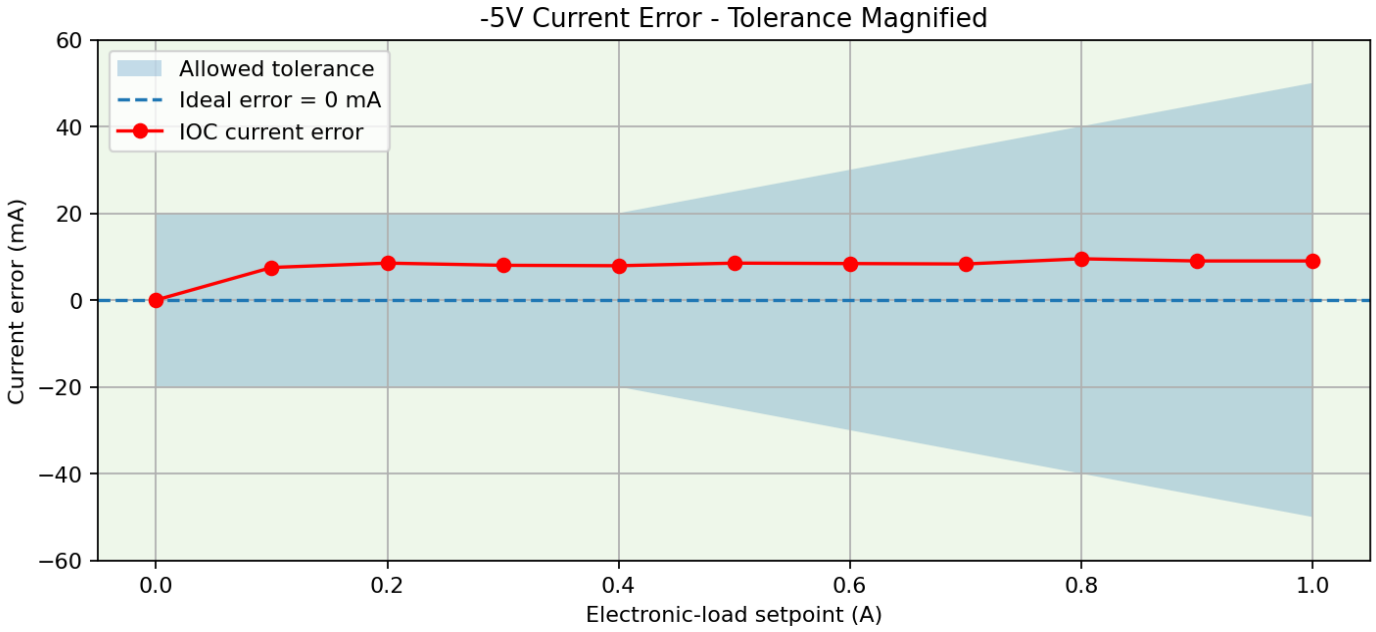
Attempt 1 — PASS IOC channel: M5V Start: 2026-10-18T18:40:59 End: 2026-10-18T18:41:28

Ideal versus measured response





Current error detail

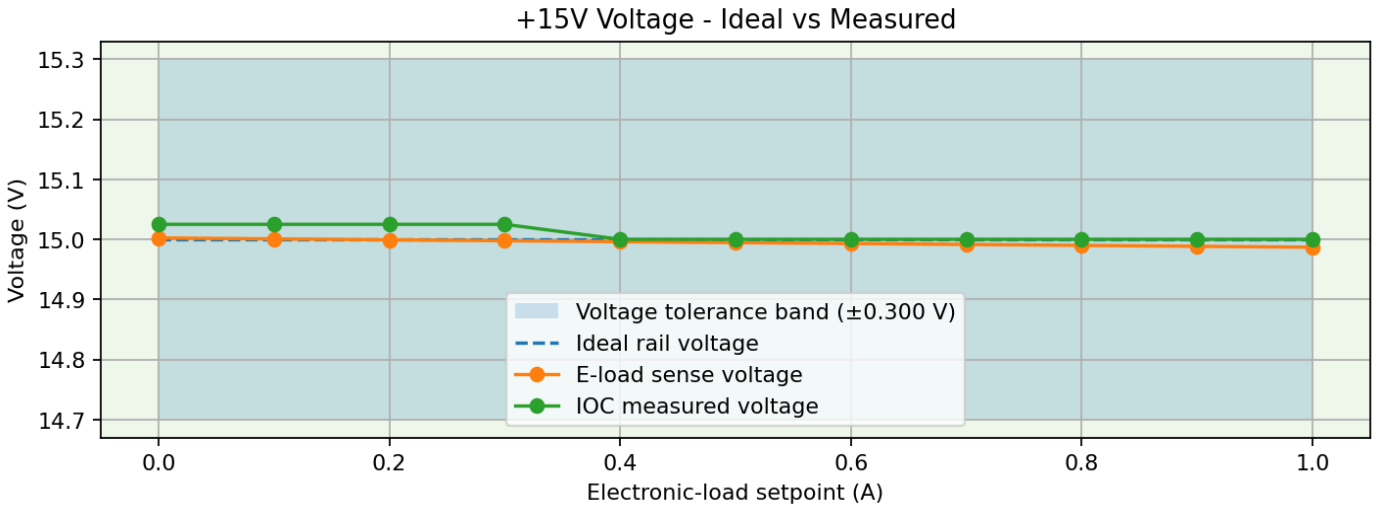


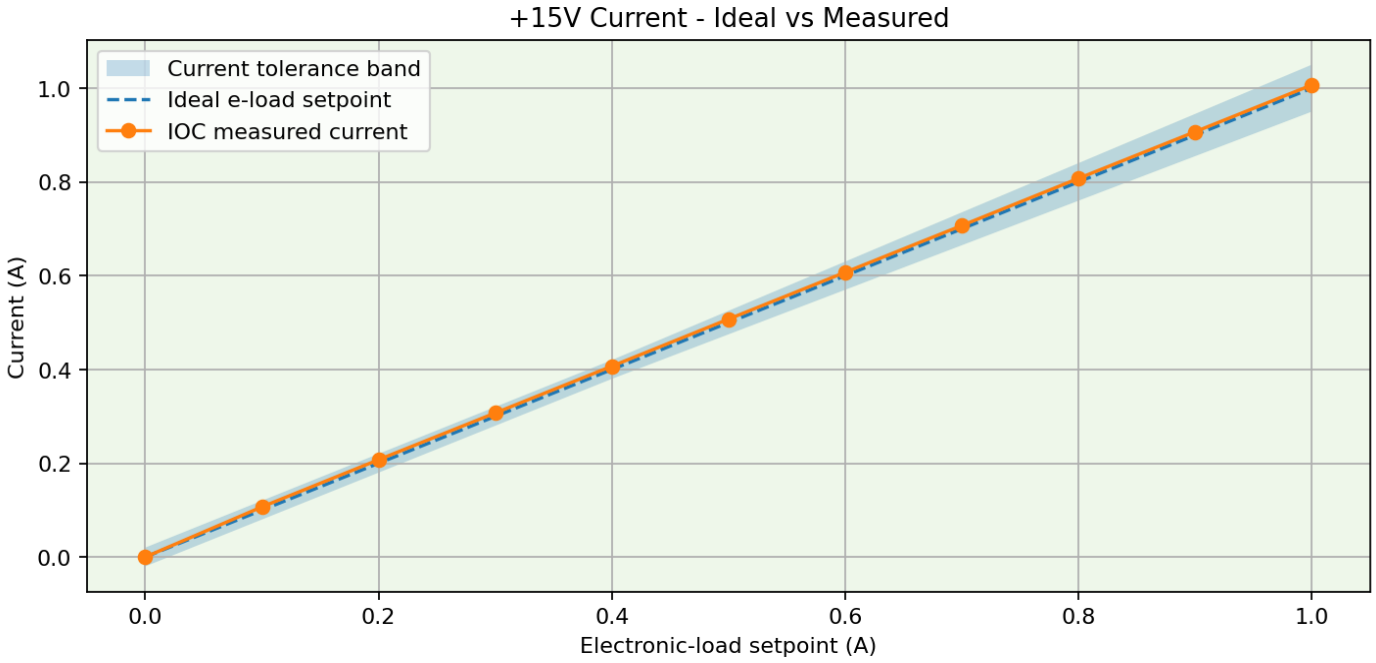
Set I (A)	Rigol V (V)	V ref (V)	IOC V (V)	V err (V)	V tol (V)	Rigol I (A)	IOC I (A)	I err (A)	I tol (A)	Result
0.0000	5.0033	-5.0033	-5.0000	+0.0033	0.1001	0.0000	0.0000	+0.0000	0.0200	PASS
0.1000	4.9990	-4.9990	-5.0000	-0.0010	0.1000	0.0986	0.1075	+0.0089	0.0200	PASS
0.2000	4.9950	-4.9950	-5.0000	-0.0050	0.1000	0.1995	0.2085	+0.0090	0.0200	PASS
0.3000	4.9911	-4.9911	-5.0000	-0.0089	0.1000	0.2993	0.3080	+0.0087	0.0200	PASS
0.4000	4.9871	-4.9871	-4.9750	+0.0121	0.1000	0.3991	0.4079	+0.0088	0.0200	PASS
0.5000	4.9832	-4.9832	-4.9750	+0.0082	0.1000	0.4997	0.5085	+0.0088	0.0250	PASS
0.6000	4.9792	-4.9792	-4.9750	+0.0042	0.1000	0.5998	0.6084	+0.0086	0.0300	PASS
0.7000	4.9753	-4.9753	-4.9750	+0.0003	0.1000	0.6996	0.7083	+0.0087	0.0350	PASS
0.8000	4.9712	-4.9712	-4.9750	-0.0038	0.1000	0.8003	0.8095	+0.0092	0.0400	PASS
0.9000	4.9675	-4.9675	-4.9750	-0.0075	0.1000	0.8998	0.9090	+0.0092	0.0450	PASS
1.0000	4.9636	-4.9636	-4.9500	+0.0136	0.1000	0.9997	1.0090	+0.0093	0.0500	PASS

+15V detailed results - latest attempt

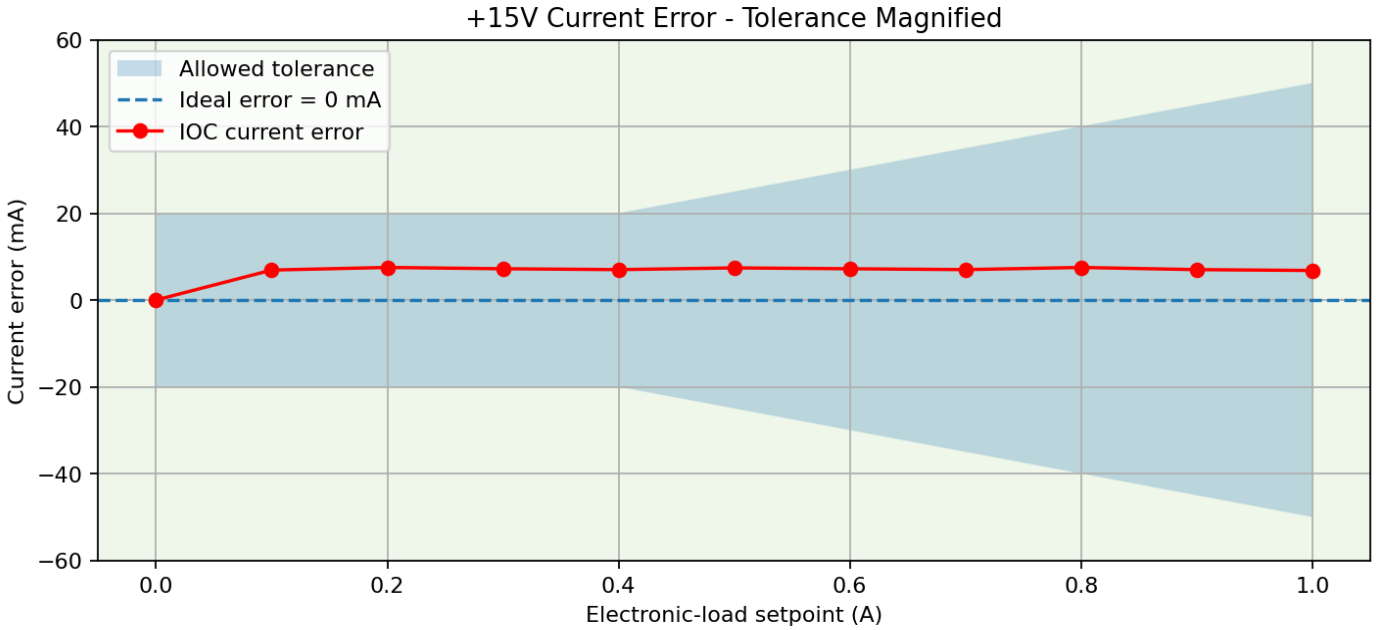
Attempt 1 — PASS IOC channel: P15V Start: 2026-10-18T18:43:04 End: 2026-10-18T18:43:32

Ideal versus measured response





Current error detail

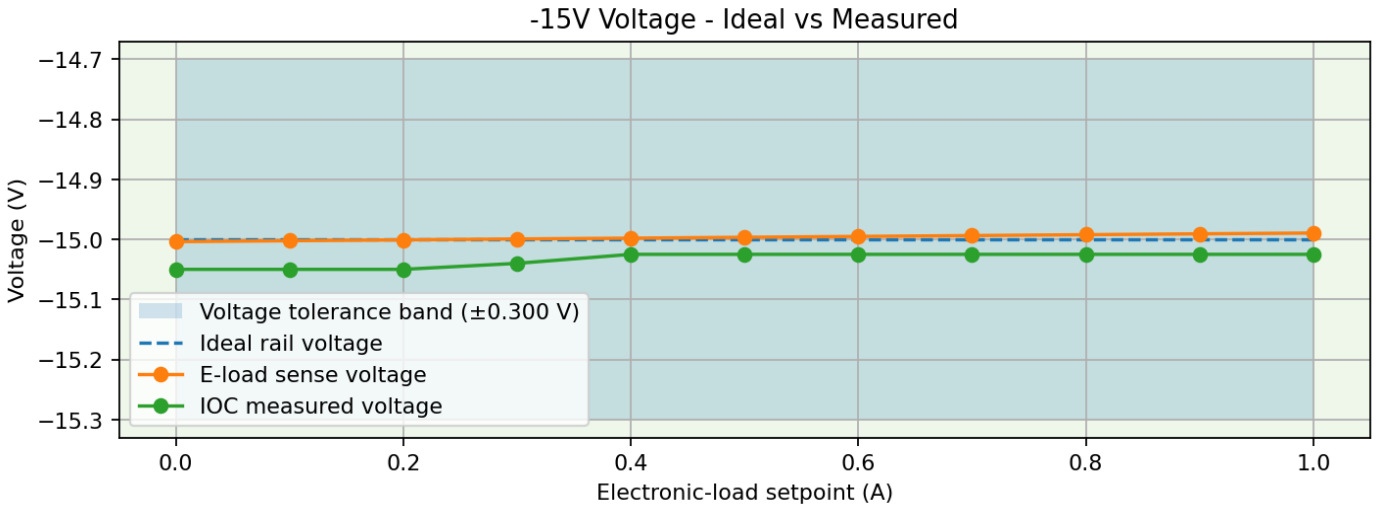


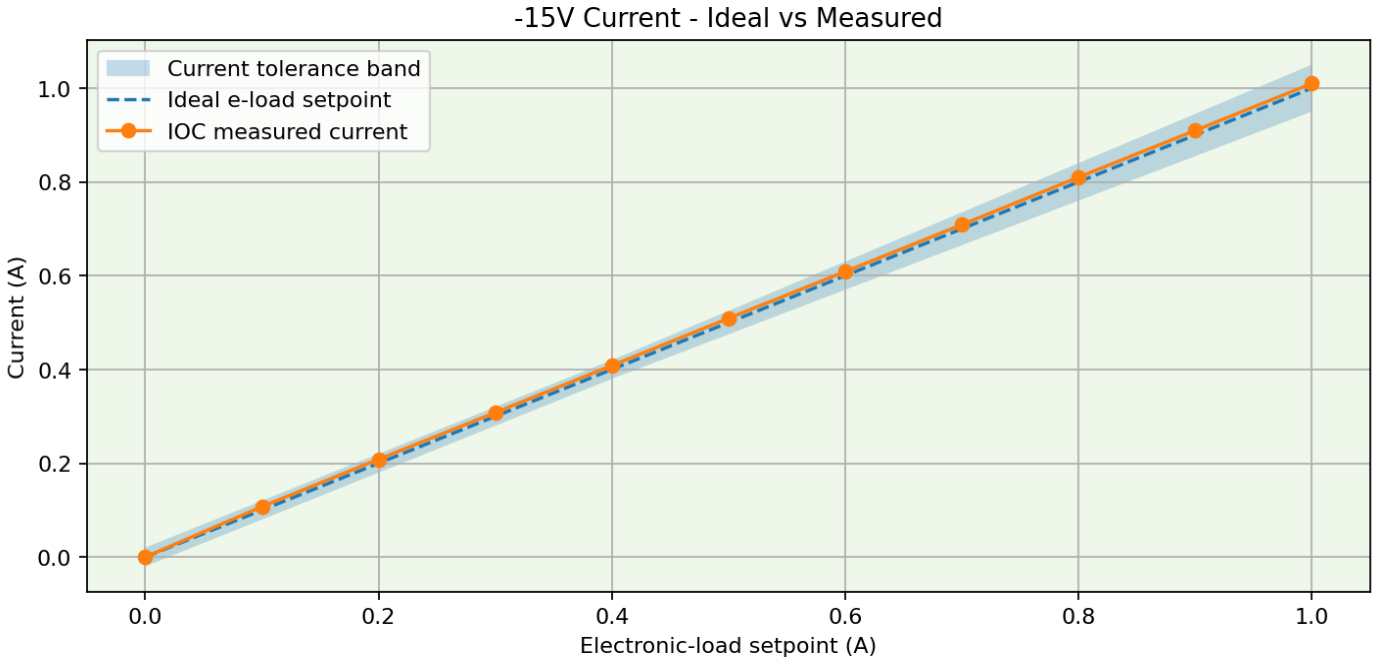
Set I (A)	Rigol V (V)	V ref (V)	IOC V (V)	V err (V)	V tol (V)	Rigol I (A)	IOC I (A)	I err (A)	I tol (A)	Result
0.0000	15.0023	+15.0023	+15.0250	+0.0227	0.3000	0.0000	0.0000	+0.0000	0.0200	PASS
0.1000	15.0007	+15.0007	+15.0250	+0.0243	0.3000	0.0985	0.1069	+0.0084	0.0200	PASS
0.2000	14.9992	+14.9992	+15.0250	+0.0258	0.3000	0.1995	0.2075	+0.0080	0.0200	PASS
0.3000	14.9977	+14.9977	+15.0250	+0.0273	0.3000	0.2993	0.3072	+0.0079	0.0200	PASS
0.4000	14.9961	+14.9961	+15.0000	+0.0039	0.2999	0.3991	0.4070	+0.0079	0.0200	PASS
0.5000	14.9945	+14.9945	+15.0000	+0.0055	0.2999	0.4997	0.5074	+0.0077	0.0250	PASS
0.6000	14.9930	+14.9930	+15.0000	+0.0070	0.2999	0.5999	0.6072	+0.0073	0.0300	PASS
0.7000	14.9915	+14.9915	+15.0000	+0.0085	0.2998	0.6996	0.7070	+0.0074	0.0350	PASS
0.8000	14.9900	+14.9900	+15.0000	+0.0100	0.2998	0.8003	0.8075	+0.0072	0.0400	PASS
0.9000	14.9884	+14.9884	+15.0000	+0.0116	0.2998	0.8999	0.9070	+0.0071	0.0450	PASS
1.0000	14.9868	+14.9868	+15.0000	+0.0132	0.2997	0.9997	1.0068	+0.0071	0.0500	PASS

-15V detailed results - latest attempt

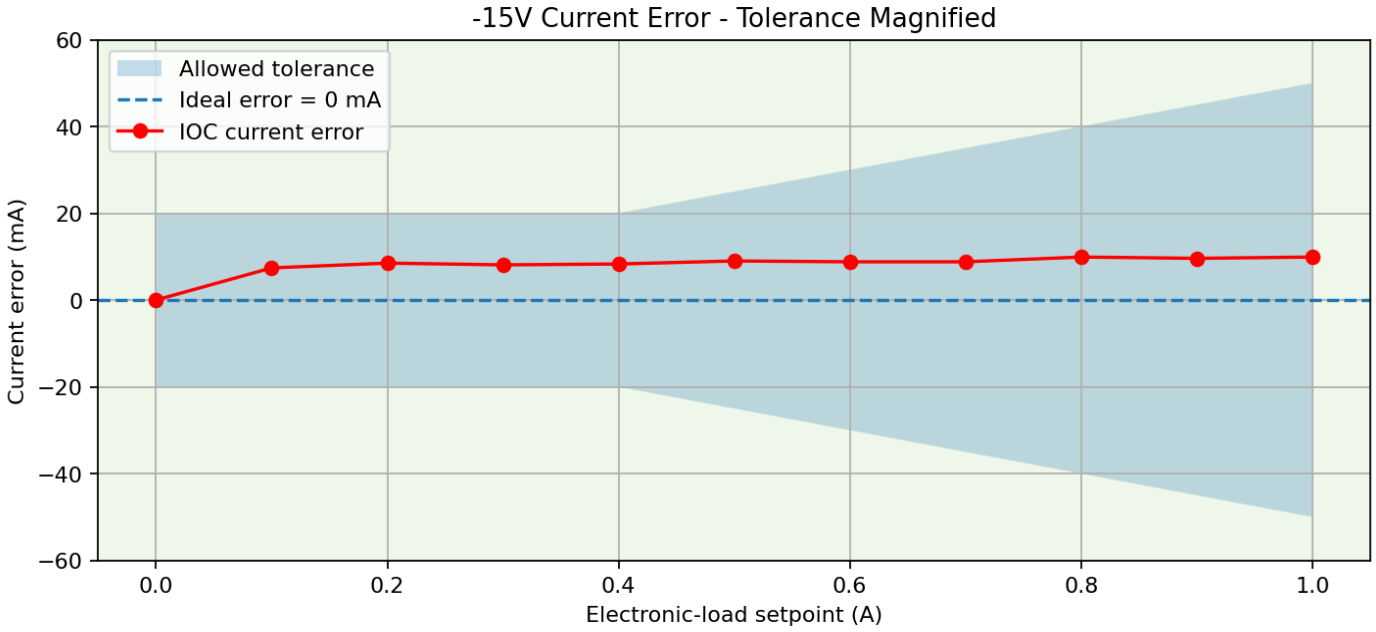
Attempt 3 — PASS IOC channel: M15V Start: 2026-10-18T18:46:38 End: 2026-10-18T18:47:06

Ideal versus measured response





Current error detail



Set I (A)	Rigol V (V)	V ref (V)	IOC V (V)	V err (V)	V tol (V)	Rigol I (A)	IOC I (A)	I err (A)	I tol (A)	Result
0.0000	15.0039	-15.0039	-15.0500	-0.0461	0.3001	0.0000	0.0000	+0.0000	0.0200	PASS
0.1000	15.0023	-15.0023	-15.0500	-0.0477	0.3000	0.0985	0.1074	+0.0089	0.0200	PASS
0.2000	15.0009	-15.0009	-15.0500	-0.0491	0.3000	0.1995	0.2085	+0.0090	0.0200	PASS
0.3000	14.9994	-14.9994	-15.0400	-0.0406	0.3000	0.2992	0.3081	+0.0089	0.0200	PASS
0.4000	14.9981	-14.9981	-15.0250	-0.0269	0.3000	0.3992	0.4083	+0.0091	0.0200	PASS
0.5000	14.9966	-14.9966	-15.0250	-0.0284	0.2999	0.4997	0.5090	+0.0093	0.0250	PASS
0.6000	14.9952	-14.9952	-15.0250	-0.0298	0.2999	0.5998	0.6088	+0.0090	0.0300	PASS
0.7000	14.9938	-14.9938	-15.0250	-0.0312	0.2999	0.6996	0.7088	+0.0092	0.0350	PASS
0.8000	14.9923	-14.9923	-15.0250	-0.0327	0.2998	0.8003	0.8099	+0.0096	0.0400	PASS
0.9000	14.9909	-14.9909	-15.0250	-0.0341	0.2998	0.8999	0.9096	+0.0097	0.0450	PASS
1.0000	14.9895	-14.9895	-15.0250	-0.0355	0.2998	0.9997	1.0099	+0.0102	0.0500	PASS

System test checklist — merged automatic/manual evidence

Each checklist step is filled from automatic rail-test evidence whenever the software can prove it. Only items that require visual/operator judgment remain manual. The source column identifies whether the current result came from automatic evidence, the latest checklist entry, or is still pending.

ST-01: Power On and Initialization — PASS

Method: Apply power to the complete system and verify Arduino, display, sensor channels, and network operation.
Pass criterion: The system initializes properly and all six channels are operational.

#	Checklist item	Result	Source / evidence
1	All power, sensor, display, and Ethernet connections have been inspected.	PASS	CHECKLIST
2	The electronic load is disabled before power-up.	PASS	CHECKLIST
3	The required monitored power supplies are energized.	PASS	CHECKLIST
4	Power has been applied to the Arduino and monitoring electronics.	PASS	CHECKLIST
5	The Arduino startup sequence completed normally.	PASS	CHECKLIST
6	The GIGA Display initialized.	PASS	AUTO: The operator interacted with the initialized GIGA Display during rail testing.
7	All six expected rail channels are displayed.	PASS	AUTO: Confirmed during the latest automated test of all six rails.
8	The EPICS IOC is running.	PASS	AUTO: A completed rail test required a running EPICS IOC.
9	The IOC established communication with the monitor.	PASS	AUTO: A completed rail test required successful IOC/monitor communication.
10	No startup error or unexpected behavior requiring manual recovery was observed.	PASS	CHECKLIST

ST-02: Rail Identification — PASS

Method: Correlate each displayed channel with its physical rail connection and polarity.
Pass criterion: The correct rail identification and polarity are shown for each connected rail.

#	Checklist item	Result	Source / evidence
1	All six monitored rails are energized and connected to their intended monitor inputs.	PASS	AUTO: Confirmed during the latest automated test of all six rails.
2	Each physical rail connection was correlated with the corresponding displayed channel.	PASS	AUTO: Confirmed during the latest automated test of all six rails.
3	The +5VA, +5VB, +5V, -5V, +15V, and -15V rail identifications are correct.	PASS	AUTO: Confirmed during the latest automated test of all six rails.
4	Positive and negative rail voltage polarity is reported correctly.	PASS	AUTO: Recorded IOC voltage data has the correct polarity on all six latest rail tests.
5	Negative-rail current follows the sign convention used by the firmware.	PASS	AUTO: Both negative-rail IOC currents follow the positive load-current magnitude convention.

ST-03: Voltage Accuracy — PASS

Method: Measure each rail with the reference instrument and compare it with the EPICS monitor value at multiple operating points.
Pass criterion: Voltage error is within ±2% across the full tested operating range.

#	Checklist item	Result	Source / evidence
1	All six rails have a completed latest voltage-measurement attempt.	PASS	AUTO: All six latest rail attempts satisfy the configured voltage acceptance criterion.
2	Every required voltage sample in the latest attempts meets the ±2% system-level criterion.	PASS	AUTO: All six latest rail attempts satisfy the configured voltage acceptance criterion.

ST-04: Current Accuracy — PASS

Method: Apply multiple electronic-load currents to each rail and compare the reference current with the monitor measurement.
Pass criterion: Current error is within ±5% across the full intended operating range; reference uncertainty/resolution is considered at low current.

#	Checklist item	Result	Source / evidence
1	All six rails have a completed latest current-measurement attempt.	PASS	AUTO: All six latest rail attempts satisfy the configured current acceptance criterion.
2	Every required current sample in the latest attempts meets the ±5% system-level criterion, with the configured low-current floor.	PASS	AUTO: All six latest rail attempts satisfy the configured current acceptance criterion.

ST-05: Dynamic Load Response — PASS

Method: Change electronic-load current while monitoring the local display and EPICS values.
Pass criterion: Current and voltage readings change according to the load change.

#	Checklist item	Result	Source / evidence
1	The electronic-load current was changed over a safe range while the system was operating.	PASS	AUTO: Automated current sweeps were recorded on: +5VA, +5VB, +5V, -5V, +15V, -15V.
2	The local display current reading changed in response to the load change.	PASS	AUTO: Confirmed during rail test(s): +5VA, +5VB, +5V, -5V, +15V, -15V
3	The EPICS current reading changed in response to the load change.	PASS	AUTO: EPICS current followed the programmed load sweep on: +5VA, +5VB, +5V, -5V, +15V, -15V.
4	The local and EPICS voltage readings responded appropriately to any rail-voltage change under load.	PASS	AUTO: Confirmed during rail test(s): +5VA, +5VB, +5V, -5V, +15V, -15V
5	No channel became unresponsive or produced an intermittent communication failure during the load change.	PASS	AUTO: All six latest rail tests completed without a Channel Access read failure.

ST-06: Local Display — PASS

Method: Observe voltage, current, rail identification, limits, and status for all channels.
Pass criterion: All necessary information is visible and correlates with the connected rail.

#	Checklist item	Result	Source / evidence
1	All six rail identifications are visible on the GIGA Display.	PASS	AUTO: Confirmed during the latest automated test of all six rails.
2	Voltage is visible for each rail and corresponds with the connected rail.	PASS	AUTO: Confirmed during the latest automated test of all six rails.
3	Current is visible for each rail and corresponds with the connected rail.	PASS	AUTO: Confirmed during the latest automated test of all six rails.
4	Configured limits/status indications are visible for each rail.	PASS	AUTO: Confirmed during the latest automated test of all six rails.
5	Communication/network state is visible and reasonable for the current operating condition.	PASS	AUTO: Confirmed during the latest automated test of all six rails.

ST-07: EPICS Communications — PASS

Method: Run the IOC and verify all required voltage, current, and status process variables.
Pass criterion: Required process variables are connected and update properly; local display and EPICS agree within displayed resolution.

#	Checklist item	Result	Source / evidence
1	The complete monitor is powered and connected to Ethernet.	PASS	AUTO: All six latest rail tests completed with a live Ethernet/IOC connection.
2	The EPICS IOC is running.	PASS	AUTO: A completed rail test required the EPICS IOC to be running.
3	The IOC connects to the Arduino TCP server.	PASS	AUTO: The IOC connection check and rail PV resolution succeeded during testing.
4	Voltage and current process variables for all six rails are connected and updating.	PASS	AUTO: Voltage/current PVs for all six rails were resolved and sampled successfully.
5	EPICS values have been compared with the local display and agree within displayed resolution/rounding.	PASS	AUTO: Confirmed during the latest automated test of all six rails.
6	A load change on a selected rail caused the corresponding EPICS voltage/current values to update.	PASS	AUTO: Recorded EPICS current values followed at least one programmed load sweep.
7	The load-change/update check was completed on at least one positive rail.	PASS	AUTO: A positive rail completed a multi-point load sweep with valid EPICS updates.
8	The load-change/update check was completed on at least one negative rail.	PASS	AUTO: A negative rail completed a multi-point load sweep with valid EPICS updates.

ST-08: Limit / Alarm Operation — PASS

Method: Set a safe test limit and change load or supply until the threshold is crossed, then return inside limits.
Pass criterion: The proper limit indication is created when the threshold is crossed and clears after the value returns inside the limits.

#	Checklist item	Result	Source / evidence
1	A rail was selected for the limit/status test.	PASS	CHECKLIST
2	The existing voltage/current limit settings were recorded before the test.	PASS	CHECKLIST
3	A safe test limit was configured without exceeding the rating of the monitored supply, sensor, wiring, or electronic load.	PASS	CHECKLIST
4	Normal status was verified while the measurement remained inside the limit.	PASS	CHECKLIST
5	The applied load or supply was changed until the selected threshold was crossed.	PASS	CHECKLIST

#	Checklist item	Result	Source / evidence
6	The local status indication changed appropriately when the threshold was crossed.	PASS	CHECKLIST
7	The corresponding EPICS status changed appropriately when the threshold was crossed.	PASS	CHECKLIST
8	The measurement was returned to the normal operating range.	PASS	CHECKLIST
9	Local and EPICS status returned to normal.	PASS	CHECKLIST
10	The original limit setting was restored.	PASS	CHECKLIST

ST-09: Persistent Settings — PASS

Method: Change user-adjustable settings, power-cycle the controller/monitor electronics, and verify the settings after restart.
Pass criterion: Settings intended to be persistent are restored without Ethernet communication or manual re-entry.

#	Checklist item	Result	Source / evidence
1	One or more user-adjustable settings were changed to known values.	PASS	CHECKLIST
2	The firmware accepted the changed settings.	PASS	CHECKLIST
3	Power was removed from the Arduino and monitor electronics.	PASS	CHECKLIST
4	The system remained unpowered for several seconds.	PASS	CHECKLIST
5	Power was restored.	PASS	CHECKLIST
6	Firmware initialization completed normally after power restoration.	PASS	CHECKLIST
7	The previously modified settings were read after restart.	PASS	CHECKLIST
8	The restored settings match the values stored before power removal.	PASS	CHECKLIST

ST-10: Communication Recovery — PASS

Method: Disconnect Ethernet while the system is running, reconnect it, and monitor local operation and the IOC.
Pass criterion: Local monitoring continues during the network interruption and normal EPICS communication resumes after reconnection without firmware reprogramming.

#	Checklist item	Result	Source / evidence
1	Normal communication between the monitor and IOC was established before the interruption.	PASS	AUTO: Normal monitor/IOC communication was established during rail testing.
2	EPICS values were verified to be updating before the interruption.	PASS	AUTO: EPICS values were sampled successfully before any communication-recovery test.
3	The Ethernet cable/network connection was disconnected while the system was running.	PASS	CHECKLIST
4	Local monitoring and the GIGA Display continued operating during the Ethernet interruption.	PASS	CHECKLIST
5	The IOC response to communication loss was observed and was reasonable.	PASS	CHECKLIST
6	Ethernet was reconnected.	PASS	CHECKLIST
7	Communication was re-established without firmware reprogramming.	PASS	CHECKLIST
8	Process variables resumed updating with valid values.	PASS	CHECKLIST

ST-11: Repeatability — PASS

Method: Make repeated measurements at selected points, including after a power cycle.
Pass criterion: Measurements remain within the specified acceptance limits with no intermittent channel failures.

#	Checklist item	Result	Source / evidence
1	Repeated measurements were made at selected operating points.	PASS	AUTO: Multiple completed measurement attempts exist for: -15V, +5VB.
2	Repeated measurements were made after at least one power cycle.	N/A	CHECKLIST
3	Repeated voltage measurements remained within the specified acceptance limits.	PASS	AUTO: Repeated voltage samples remain within the configured acceptance limit.
4	Repeated current measurements remained within the specified acceptance limits.	PASS	CHECKLIST
5	No intermittent sensor or communication failures were observed during repeatability testing.	PASS	AUTO: Repeated attempts completed without an acquisition/communication abort.

ST-12: End-To-End Demonstration — PASS

Method: Operate the complete system and demonstrate local and remote monitoring while changing the load on a selected channel.
Pass criterion: Hardware, firmware, display, network, IOC, and test software operate cooperatively with no system-level failure.

#	Checklist item	Result	Source / evidence
1	The complete hardware and firmware system is operating simultaneously.	PASS	AUTO: The complete measurement chain has been exercised successfully on all six rails.
2	The local GIGA Display is operating and showing valid rail information.	PASS	AUTO: Local-display identity and update behavior were confirmed on all six latest rail tests.
3	Ethernet and the EPICS IOC are connected and updating.	PASS	AUTO: Ethernet/IOC communication was active while all six latest rail tests were acquired.
4	Remote monitoring through EPICS/Phoebus is operating.	PASS	CHECKLIST
5	The test software can acquire/record data from the system.	PASS	AUTO: The Python test software successfully acquired and saved measurement data.
6	Changing the load on a selected channel is reflected in both local and remote monitoring.	PASS	AUTO: A load sweep was reflected in recorded EPICS data and confirmed on the local display.
7	The end-to-end demonstration completed without a system-level failure.	PASS	CHECKLIST

Test completion criteria

Completion criterion	Status
Every test identified in the system test matrix has been executed.	YES
All six rail channels have completed voltage and current verification.	YES
All system-test items have an effective PASS result.	YES
Failed/incomplete tests are retained in the append-only checklist history for correction/retest or documentation as limitations.	RECORDED
Raw measurement data has been preserved.	YES
Final test report has been generated.	YES

Engineering basis for tolerances

The LTC2945 uses a 12-bit VIN converter with 25 mV/LSB and a 102.4 V full-scale. Its guaranteed total unadjusted error is ±0.75% of full-scale, which corresponds to approximately ±0.768 V for VIN. The report uses a tighter characterization/acceptance limit of 2% with a 100 mV floor, as this is a more reasonable limit at the used range.

Rail	Shunt	LTC2945 ADC TUE equivalent	1% shunt at test max	Recommended acceptance
+5VA	39.0 mΩ	19.7 mA	20.0 mA	±max(5% reading, 25 mA)
+5VB	39.0 mΩ	19.7 mA	20.0 mA	±max(5% reading, 25 mA)
+5V	50.0 mΩ	15.4 mA	10.0 mA	±max(5% reading, 20 mA)
-5V	50.0 mΩ	15.4 mA	10.0 mA	±max(5% reading, 20 mA)
+15V	50.0 mΩ	15.4 mA	10.0 mA	±max(5% reading, 20 mA)
-15V	50.0 mΩ	15.4 mA	10.0 mA	±max(5% reading, 20 mA)

The current system-test criterion is ±5% of the reference value. At low current, where a percentage-only limit approaches zero and instrument/ADC resolution becomes significant, the script uses the larger of the 5% window and the configured absolute floor. The floor also covers the LTC2945 delta-sense full-scale TUE, shunt tolerance, and a small reference/rounding margin.